



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

September 9, 2016

Name:

Surname:

Student ID:

Row:

Column:

Time: 2 hour 30 minutes

Prof. Marcello Restelli

Maximum Mark: 31

- The following exam is composed of **10 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided outside these spaces will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices** like laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Ex. 9	Ex. 10	Tot.
/ 5	/ 5	/ 5	/ 2	/ 2	/ 2	/ 2	/ 2	/ 3	/ 3	/ 31

Exercise 1 (5 points)

Describe the supervised learning technique denominated **K-Nearest Neighbor** for classification problems.

Exercise 2 (5 points)

Describe the **Bias-Variance tradeoff** for regression problems. Explain how is it possible to evaluate the bias-variance tradeoff by looking at the train error and at the test error.

Exercise 3 (5 points)

Describe the difference between **on-policy** and **off-policy** reinforcement learning techniques. Make an example of an on-policy algorithm and an example of an off-policy algorithm.

Exercise 4 (2 point)

Consider the case in which you have an estimated state/action value of $Q(s, a) = 1$ you perform an action a , gain a reward of $R_t = 2$ and reach state s' . In s' you could perform actions a'_1, a'_2, a'_3 . You previously estimated the values for these actions and you obtained $Q(s', a'_1) = 7.5$, $Q(s', a'_2) = 5$ and $Q(s', a'_3) = 10$. Assume that following the current policy π action a'_2 is chosen in state s' .

Consider a discount factor $\gamma = 0.2$ and a learning rate $\alpha = 0.5$. Tell if the following values are consistent with the use of SARSA and/or Q-learning algorithms after the update:

1. $Q(s, a) = 2.5$
2. $Q(s, a) = 2.75$
3. $Q(s, a) = 2$
4. $Q(s, a) = -1.5$

Motivate the answers you provided.

The update rules in this case of SARSA and Q-learning are, respectively:

$$\begin{aligned} Q(s, a) &\leftarrow Q(s, a) + \alpha(R_t + \gamma Q(s', a'_2) - Q(s, a)) \\ &= 1 + 0.5(1 + 0.2 \cdot 5 - 1) = 2 \end{aligned}$$

$$\begin{aligned} Q(s, a) &\leftarrow Q(s, a) + \alpha(R_t + \gamma \max_{a'} Q(s', a') - Q(s, a)) \\ &= 1 + 0.5(1 + 0.2 \cdot 10 - 1) = 2.5 \end{aligned}$$

Thus:

1. Consistent with Q-learning;
2. Inconsistent with both;
3. Consistent with SARSA;
4. Inconsistent with both.

Exercise 5 (2 point)

Provide examples of either a Bayesian or a frequentist MAB algorithm showing the following properties:

1. It provides a tight theoretical upper bound on the expected regret in the stochastic setting;
2. It provides a tight theoretical upper bound on the expected regret in the adversarial setting;
3. It modifies the statistics of all the available arms at each round;
4. It incorporates *a priori* knowledge about the arms.

Motivate your answers.

1. UCB1, as well as Thompson sampling, is order optimal and presents a theoretical upper bound over the expected regret of order $O(\log(T))$, where T is the horizon of the bandit process;
2. EXP3 is order optimal for the adversarial MAB problem, with a theoretical upper bound over the (adversarial) regret of order $O(\sqrt{T})$;
3. UCB1 has to update the upper bound of all the arms, since there is the term $\log(t)$ which changes at each turn;
4. Thompson sampling is able to incorporate *a priori* information at the beginning of the process by means of the prior we instantiate.

Exercise 6 (2 point)

Tell if the following statements about linear regression are true or false. Justify your choices.

1. The use of a linear model usually suffice to capture all the characteristics of the considered regression problem;
2. The use of a nonlinear combination of the original features in a linear model can be used to capture nonlinear effects;
3. Linear regression may outperform more complex regression techniques, especially if the data are noisy;
4. A closed form solution for linear regression is available.

1. FALSE: the linear model are useful to understand the behavior of a process and chose the right technique to use. Usually real-world problems presents naturally non-linear characteristics;
2. TRUE: a linear model is able to incorporate non-linearity by means of base functions $\phi_i(x)$, which generally could have arbitrary form, thus also non-linear combination of the original features. The model still remains linear in its parameters;
3. TRUE: in the case of high noise presence the use of simple models might contain the variance of the model, even if it might introduce some bias. In this case, more complex models might overfit by trying to model also the noise;
4. TRUE: if we use the Least Square method we have a closed form solution for the parameters w :

$$w = (\Phi^T \Phi)^{-1} \Phi^T t$$

where, Φ is the design matrix and t is a vector containing the true targets.

Exercise 7 (2 point)

Which of the following statements are true?

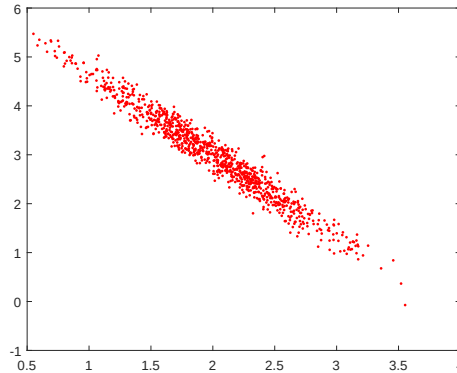
1. Suppose you have $2D$ input samples (i.e., $x_i \in \mathbb{R}^2$). The decision boundary of an SVM (with a linear kernel) is a straight line;
2. If you train a multi-class SVM with the one-vs-all method, it is not possible to use a kernel;
3. The maximum value of the Gaussian kernel is 1;
4. If the data are linearly separable, an SVM using a linear kernel will return the same parameters w regardless of the chosen regularization value C .

Motivate your answers.

1. TRUE: in $\mathbb{R}^2$ the boundary of a given parameter vector w is the set of points $x \in \mathbb{R}^2$ s.t. $w^T x + b = 0$, which is the equation for a line;
2. FALSE: there is no correlation between the use of kernel and the use of methodologies for dealing with more than two classes;
3. TRUE: $\max_{x \in \mathbb{R}} e^{-x^2} = 1$;
4. FALSE: the regularization parameter C is the responsible of the between-margins zone thickness, thus even if the two classes are linearly separable it might influence completely both the margins and the boundary.

Exercise 8 (2 point)

Consider the following dataset:



1. Draw the directions of the principal components and provide an approximate and consistent guess of the values of the loadings.
2. Are they unique?

Motivate adequately your answers.

1. The direction of the principal directions (or loadings) are the ones where the data are more spread. In order to be consistent they should be orthogonal and normal vectors. Since the data lies in a rectangle of approximately 2 units in horizontally and 4 vertically, the data have approximately the direction of the angle of $\frac{5\pi}{6}$, thus the directions of the first w_1 and second w_2 principal directions are:

$$w_1 = \begin{bmatrix} -\frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} \quad w_2 = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$$

2. Even if the procedure to compute the principal component is deterministic, they are not unique, since the choice of a combination of $w_1, w_2, -w_1, w_2, w_1, -w_2$ or $-w_1, -w_2$ is a feasible choice.

Exercise 9 (3 points)

Let us assume that the solution $\hat{w} = [w_0 w_1]$ found by the Least Square method for a regression problem has result:

$$\hat{t} = w_0 + w_1 x = 1 + 2x$$

Now assume we would like to repeat the same regression with a Gaussian Bayesian prior over the parameter space $[w_0; w_1]$ with mean $w_p = [4; 2]$ and covariance matrix $\sigma_2 = I_2$, where I_2 is the identity matrix of order 2. Which of the following are consistent solutions to the previous regression problem with the previously specified Bayesian prior?

1. $w = [3; 2];$
2. $w = [1; 2];$
3. $w = [2; 7];$
4. $w = [4; 3];$
5. $w = [6; 2];$
6. $w = [0; 1].$

Motivate your answer adequately. Hint: the use of a figure might help.

Since here we are considering the Least Square method, the only feasible points for the posterior of the parameter vector are lying on the segment between $[1; 2]$ and $[4; 2]$ (excluding the extreme points which might occur if we have infinite or zero data, respectively).

This means that the only feasible answer is the number 1; all the others are inconsistent.

Exercise 10 (3 points)

Consider a linear two-class SVM classifier defined by the parameters $w = [3; 2]$ and $b = 2$. Answer the following questions providing adequate motivations.

1. How the point $x_1 = [3; 1]$ is classified according to the trained SVM?
2. How the point $x_2 = [-4; 1]$ is classified according to the trained SVM?
3. Is the SVM providing a correct prediction if the point $x_3 = [3; -1]$ is classified in the positive class?
4. Assume to collect a new sample $x_4 = [-2; 2]$ with positive label. Do you need to retrain the SVM?
5. Does the answer to the previous question apply to a generic new sample?
6. Give an example of a point which is on the boundary of the SVM.

1. $\text{sgn}(w^T x_1 + b) = \text{sgn}(3 \cdot 3 + 2 \cdot 1 + 2) = \text{sgn}(13) = 1 \rightarrow$ positive class;
2. $\text{sgn}(w^T x_2 + b) = \text{sgn}(3 \cdot (-4) + 2 \cdot 1 + 2) = \text{sgn}(-8) = -1 \rightarrow$ negative class;
3. $\text{sgn}(w^T x_3 + b) = \text{sgn}(3 \cdot 3 + 2 \cdot (-1) + 2) = \text{sgn}(9) = 1 \rightarrow$ positive class $\rightarrow$ correctly classified;
4. $w^T x_4 + b = -2 \cdot 3 + 2 \cdot 2 + 2 = 0 \rightarrow$ the point is on the boundary, thus we need to retrain;
5. In general, if the new point is between or on the margins (i.e., would have been a support vector) we need to retrain. Moreover, if this is not the case but the point would have been classified in a wrong way, we have to retrain. Finally if non of the above hold, we do not need to retrain;
6. x_4 since $w^T x_4 + b = 0$